

CYBER SECURITY MINOR

The minor requires students to have a program of study in computer engineering, computer science, software engineering, or management information systems with the goal of enabling them to work in cyber security. The minor consists of a series of lab based courses that are designed to provide students with both the technical background and the hands-on experiences along with the theoretical background to allow them to compete for jobs in cyber security.

Objectives

The minor in cyber security is designed to prepare students with the technical skills for entry into cybersecurity positions in industry or government agencies.

A few years after graduation, students completing the cyber security minor should be:

1. Contributing to their communities and society in the area of cyber security technology and applications and demonstrating an understanding of contemporary security issues, both technological and societal.
2. Advancing in their careers through application of their knowledge of cyber security
3. Working effectively as team members and demonstrating ethics and responsible behavior
4. Applying cyber security methods and concepts to the general area of their BS degree
5. Continuing their professional development through life-long learning

Learning Outcomes

After earning the minor in cyber security students will

1. Demonstrate the ability to apply knowledge of cyber security concepts, tools and technologies to computer systems.
2. Understand cyber security risks, threats and countermeasures and apply this understanding to develop cyber defense strategies.
3. Demonstrate the ability to design cyber security systems to meet organizational needs within realistic constraints such as economic, environmental, social, and ethical expectations.
4. Demonstrate the ability to function on teams.

The minor requires 15 credits, including at least 6 credits in courses numbered 3000 or above taken at Iowa State University. The minor must include at least 3 credits that are not used to meet any other department, college, or university requirement. Below is the list of courses used in the minor.

Core

CYBE 2300	Cyber Security Fundamentals	3
CYBE 2310	Cyber Security Concepts and Tools	3
CYBE 3310	Application of Cryptographic Concepts to Cyber Security	3
Plus one of the following (3-4 cr.)		3
CPRE 3080	Operating Systems: Principles and Practice	
COMS 2520	Linux Operating System Essentials	
COMS 3520	Introduction to Operating Systems	
One Elective Course ¹		3
Total Credits		15

1. From department approved lists.